

Run it — investigation record

Name	Group	Class	Date
Our question			
What we are CHANGING			
What we are MEASURING			
Time we started			

Before you get any equipment Your group says out loud what could cause harm in YOUR method, and what you are doing about each one. Not the class rules — yours. No equipment until the teacher has signed you off.

1 • What is in each jar

Jar	Treatment	Anything different about this one?
1		
2		
3		

2 • Readings

Rate each jar 0 to 4 against the scale. Rate all three SEPARATELY — do not average them in your head. Photograph from the same angle every time.

Time	Jar 1	Jar 2	Jar 3	Photo taken	Anything you notice — write it NOW
Start — 0 min					
10 min					
20 min					
30 min					
NEXT WEEK					

3 • The front bench — temperature

One jar at the front has a thermometer in it and untreated steel wool. Nothing is heating it.

	0 min	5 min	10 min	15 min
Temperature (°C)				

Nothing was heated. So where is the warmth coming from?

4 • Did anything go wrong?

This is worth marks A method that failed and was honestly described is worth MORE than a tidy result you cannot explain. Write what happened, when, and what you did about it.

When	What happened, and what we did about it

5 • Planning the report

Next week you write to the client. Work out now what you are missing.

Based on what you have seen so far, what are you going to recommend?

What is the STRONGEST piece of evidence you have for that?

What is the WEAKEST part of your method — the bit they could fairly attack?

What do you still need before you can write it?

6 • Who is doing what next week

Name	Bringing / answering which challenge question

Hand this in before you leave You get it back next week. If it goes in your bag, it will not be there when you need it.

Role cards — cut into four

Every job matters and none of them depends on being the fastest writer. Swap roles at the 20-minute reading if you want to.

EQUIPMENT • Collects everything on the list and checks it against the plan. • Checks jars for chips before they are used. • Packs everything away and wipes the bench.	SET-UP • Times the vinegar soak — two minutes, then squeeze out hard. • Makes sure all three pieces are the same size. • Applies the same amount of treatment each time.
RECORDER • Writes every reading down as it happens, not afterwards. • Writes anything odd in the notes column straight away. • Makes sure the sheet is handed in at the end.	OBSERVER • Takes the photograph at every reading, from the same angle. • Rates all three jars separately against the scale. • Says out loud if one jar looks different from the other two.

The swap — somebody else’s data

Name	Group	Date
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You were not there This is the other class’s data. You do not know how they did it. They tested something you did not, and you tested something they did not — neither class can answer the client’s question alone. But before you use their numbers, you have to decide whether you trust them.

WRITE THESE DOWN BEFORE YOU DISCUSS ANYTHING. Once someone says what they think out loud, it gets very hard to notice anything different.

1 • Two things you CANNOT tell from this data

What did they not write down that you would need to know?

2 • One thing you think may have gone wrong

Be specific. Name the step.

3 • One question you would ask them

Things worth looking for Did they dry the nails the same way? Did every jar start on the same day? Were the jars in the same place — same light, same temperature? Same size nails? Is one of their three jars wildly different from the other two, and did they say why? Did they record to two decimal places or round? Is anything missing?

Now turn it around

What would the OTHER CLASS say about OUR method? Name two things they could fairly attack.

1	
2	

Why this matters

- Could either class have answered the client’s question on its own? Explain.

- Real scientists pull each other’s methods apart constantly. Why would that make science BETTER rather than worse?

The point Scientists publish their methods precisely so other people can attack them. A finding that survives other people trying to pull it apart is worth far more than one nobody checked. That is not scientists being difficult with each other — it is the part that makes the answer trustworthy.

Coming up next You are about to write your own investigation plan, and another group is about to do exactly this to it. What would YOU attack?